

Physical vs. Displayed Color

A visual color-reproduction comparison

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OBJECTIVE

Document how colors in a physical ColorChecker and two paint samples change when photographed, displayed on a laptop, and viewed beside the originals under controlled illumination.



Figure 1. Original capture containing the ColorChecker and two paint samples.

CAPTURE SETUP

iPhone SE (3rd generation), rear camera; auto exposure and white balance; 1/120 s, f/1.8, ISO 25; 4.0 mm focal length (28 mm equivalent).

WORKFLOW

Preparing the comparison images

The source photograph was separated into a ColorChecker crop and a two-sample crop. The chart was resized to 1125 x 800 pixels, and the sample image was resized to 225 x 300 pixels for display and visual comparison.

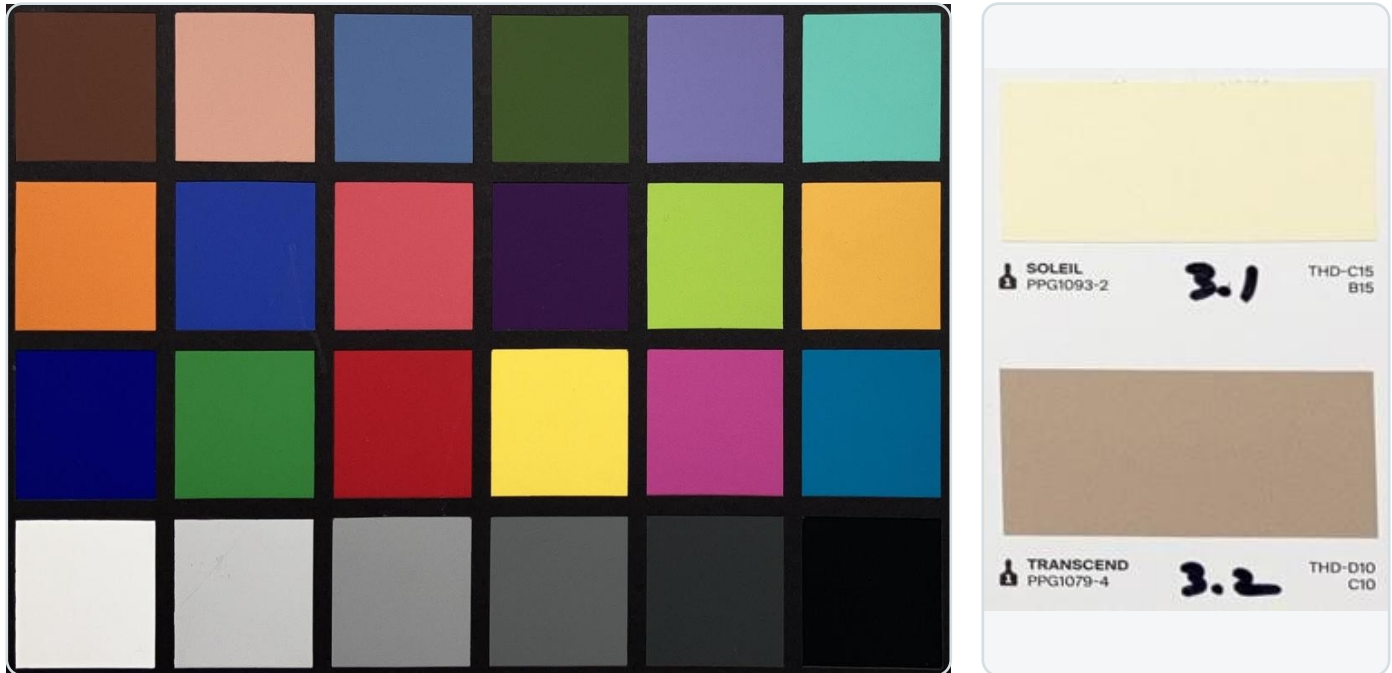


Figure 2. Cropped ColorChecker and paint samples used for display.

MATLAB IMAGE-PREPARATION WORKFLOW

```
img = imread('../data/SE03_01.jpg');  
colorChart = imcrop(img);  
samples = imcrop(img);  
colorChart = imresize(colorChart, [800 1125]);  
samples = imresize(samples, [300 225]);  
imwrite(colorChart, '../results/chart.jpg');  
imwrite(samples, '../results/samples.jpg');
```

COMPARISON

Viewing physical and displayed colors

The physical targets and their displayed images were viewed side by side in the same light booth. Observations focused on perceived lightness, hue direction, and saturation rather than numerical color difference.



Figure 3. Physical ColorChecker at left and displayed reproduction at right.

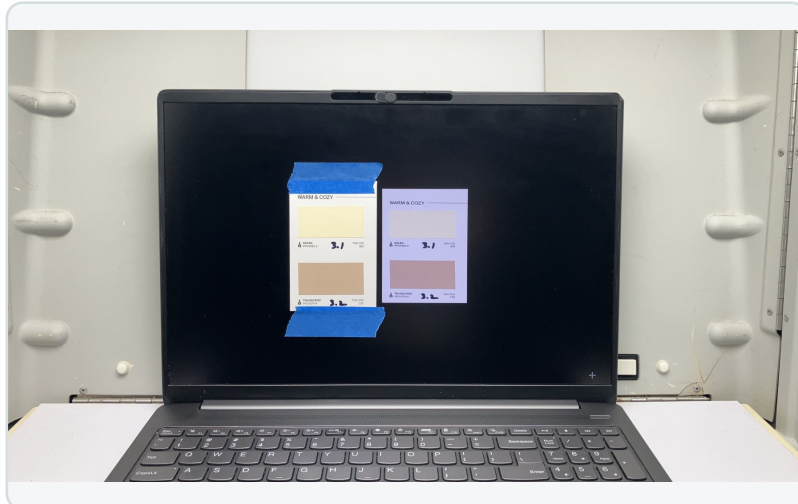


Figure 4. Physical paint samples at left and displayed reproduction at right.

KEY FINDINGS

- Many displayed patches appeared darker than their physical counterparts.
- Blue shifts and reduced yellow appearance were recurring hue differences.
- Several saturated colors lost saturation; a few became more saturated.
- Purplish blue showed the strongest reported hue mismatch.
- Blue flower was the closest visual match among the chart patches.

INTERPRETATION

The visible differences demonstrate that an uncharacterized camera-display pipeline does not preserve color appearance consistently. This observation motivated the later quantitative measurement and device-characterization work in the project sequence.

RESULTS

Detailed visual observations

Descriptions compare each displayed patch with its physical counterpart under the viewing conditions shown on the preceding page.

Patch	Patch name	Displayed appearance
1	Dark skin	Darker, redder, more saturated
2	Light skin	Lighter, bluer, less saturated
3	Blue sky	Darker, bluer, more saturated
4	Foliage	Darker, less yellow, more saturated
5	Blue flower	Very close; slightly darker
6	Bluish green	Darker, bluer
7	Orange	Lighter, less saturated
8	Purplish blue	Much bluer; very different hue
9	Moderate red	Less yellow
10	Purple	Darker
11	Yellow green	Lighter, less yellow, less saturated
12	Orange yellow	Less saturated
13	Blue	Darker, bluer
14	Green	Lighter, less yellow
15	Red	Darker
16	Yellow	Less saturated
17	Magenta	Bluer
18	Cyan	Darker, bluer
19	White (.05)	Lighter, less yellow
20	Neutral 8	Lighter, less yellow tinge
21	Neutral 6.5	Lighter, less yellow tinge
22	Neutral 5	Slightly darker
23	Neutral 3.5	Darker, less yellow tinge
24	Black	Darker, less yellow tinge
-	Patch 3.1	Slightly bluer, less saturated
-	Patch 3.2	Darker